

Assignment: House Rent Data Analysis

Objective:

This assignment aims to help students understand real-world data analysis using house rent data. Students will perform data cleaning, exploratory data analysis, visualization, and derive insights using Python.

Dataset Description:

Assume a house rent dataset with the following columns:

- City
- Area (sqft)
- Number of Bedrooms (BHK)
- Furnishing Status (Furnished, Semi-Furnished, Unfurnished)
- Tenant Preferred (Family, Bachelor, Any)
- Floor
- Total Floors
- Area Type (Built Area, Super Area, Carpet Area)
- Availability Date
- Monthly Rent
- Deposit Amount

Assignment Tasks:

Part 1: Data Understanding & Cleaning

1. Load the dataset using pandas.
2. Display the first 10 rows and summarize the dataset.

3. Identify missing values and handle them appropriately.
4. Detect outliers in rent and area columns.
5. Convert data types where necessary.

Part 2: Exploratory Data Analysis (EDA)

1. Calculate summary statistics for rent, area, and deposit.
2. Analyze average rent by city.
3. Compare rent based on furnishing status.
4. Relationship between area (sqft) and rent.
5. Rent distribution for different BHK categories.

Part 3: Data Visualization

1. Histogram of Monthly Rent.
2. Boxplot of Rent by City.
3. Bar chart for Average Rent by Furnishing Status.
4. Scatter plot of Area vs Rent.
5. Heatmap showing correlation between numerical features.

Part 4: Insights & Interpretation

1. Which city has the highest and lowest average rent?
2. Does furnishing significantly impact rent?
3. How does house size influence rent?
4. Are higher deposits always linked to higher rent?
5. Identify at least 3 business or consumer insights.

Part 5: Advanced Analysis (Optional – Bonus)

1. Build a simple linear regression model to predict rent.
2. Evaluate model performance.
3. Explain feature importance.

Deliverables:

1. Python notebook (.ipynb)
2. Word/PDF report including:
 - Problem statement
 - Methodology
 - Visualizations
 - Insights & conclusions

Evaluation Criteria:

- Data Cleaning: 20%
- Analysis & Visualization: 40%
- Insights & Interpretation: 25%
- Code Quality & Presentation: 15%

Submission Deadline:

(To be announced)

Note:

Use libraries such as pandas, numpy, matplotlib, seaborn, and scikit-learn.